LLMs and Formal Methods: A survey

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1 Scope and structure

This survey is organised around the two-way intersection between large language models and formal methods: *formal methods for constraining or checking AI-assisted software development*, and *LLMs for lowering the cost of formal specification, verification, and mechanised proof.* After a short section of surveys and roadmaps, the material is organised directly by technical category, mixing papers, systems, benchmarks, and industry artefacts where that gives the clearest picture of the space. The main emphasis is on systems with a direct formal-methods component, but the survey also notes adjacent specification and programming artefacts when they illuminate likely interfaces between natural-language intent, intermediate specifications, AI-native implementation languages, and verifier-oriented workflows. Representative surveys and roadmaps that frame the field include the main autoformalisation survey, the software-requirements survey, the theorem-proving survey, and broader roadmaps on trustworthy AI agents and formal mathematical reasoning. [1, 2, 3, 4, 5, 6]

2 Taxonomy

Area	What sits here	Section
Surveys and roadmaps	High-level surveys, position papers, and roadmaps that frame the field and its major research directions.	§3
Specification languages and requirements	Natural language to logic/specification, intent formalisation, and auditable requirement capture.	§4
Verified code generation and proof synthesis for code	Verifier-in-the-loop generation of code, contracts, assertions, and proofs for Dafny, Verus, ACSL, SPARK, and related systems.	§5
AI-assisted mechanised proving	Interactive theorem proving, premise selection, tactic generation, whole-proof synthesis, and proof repair in Lean, Isabelle, Rocq/Coq, and Metamath.	§6
Temporal logic, TLA+, model checking, and hardware FV	Temporal-logic generation, TLAPS/TLA+ proof support, and assertion synthesis for hardware formal verification.	§7
Benchmarks and datasets	Shared evaluation sets for theorem proving, verifiable code generation, repository-scale proof, TLA+, and hardware FV.	§8
Adjacent specification and AI-native coding systems	Spec-centric languages, AI-oriented implementation languages, and spec-driven development toolchains that are not themselves formal methods results but may shape how intent is structured before verification.	§9

3 Surveys and roadmaps

- **Autoformalisation survey**. [1]
- **Survey on LLMs for software-requirement formalisation**. [2]
- **Survey on deep learning for theorem proving**. [3]
- **Roadmap on LLMs and formal methods for trustworthy agents**. [4]
- **Formal mathematical reasoning as an AI frontier**. [5]
- **Two-way roadmap between requirements engineering and LLMs**. [6]

4 Specification languages and requirements

- **VERIFAI**. Early work on traceability and verification of natural-language requirements, useful as background for NL-to-spec workflows. [7]

- **Formal Specifications from Natural Language**. Direct NL-to-formal-spec framing paper. [8]
- **Autoformalization with Large Language Models**. Early LLM-based autoformalisation of informal mathematical text. [9]
- **Process-Driven Autoformalization in Lean 4**. Systematic pipeline for translating informal mathematical text into Lean 4 proofs with process supervision. [10]
- **Improving Autoformalization using Type Checking**. Shows that type-checking feedback significantly reduces autoformalization errors. [11]
- **nl2spec**. Interactive translation from unstructured requirements to temporal logic with editable sub-translations. [12]
- **NL2TL**. A general framework for mapping natural-language instructions to temporal logic across multiple domains. [13]
- **SpecGen**. Automated generation of formal specifications from natural-language requirements. [14]
- **Specify What?** and **Seeking Specifications**. Neuro-symbolic work on deriving and refining specifications from informal descriptions. [15, 16]
- **Doc2Spec**. A multi-agent approach that induces formal specifications from natural-language documents. [17]
- **Trustworthy Formal Natural Language Specifications**. A proof-assistant-native route to formalising natural-language specifications inside Lean. [18]
- **Microsoft intent formalisation agenda**. Industry framing that treats user-intent formalisation as a central reliability problem for AI-assisted coding. [19]
- **nl2postcond** and follow-on evaluation work. Microsoft-led efforts on turning programmer intent into checkable postconditions and evaluating this route for verification-aware languages. [20, 21]
- **Supporting Software Formal Verification with Large Language Models: An Experimental Study**. Broader evidence on where LLM assistance helps in formal-specification and verification workflows. [22]

5 Verified code generation and proof synthesis for code

- **DafnySynth / dafny-annotator**. LLM-assisted synthesis of Dafny assertions, invariants, and proof-oriented annotations. [23]
- **AI-assisted Dafny synthesis (Misu et al.)**. Systematic study of LLM-assisted synthesis of verified Dafny methods, evaluating multiple LLMs on generation and annotation tasks at FSE 2024. [24]
- **Laurel**. Uses LLMs to unblock automated verification by synthesising useful proof and specification steps. [25]
- **Verification with Ada/SPARK**. Evaluation of LLM-generated code in a SPARK-based verification setting. [26]

- **LLM code generation with formal specifications and reactive synthesis**. Couples code generation with formal specifications and synthesis constraints. [27]
- **VeCoGen**. Formally verified C generation using LLMs plus a verifier-guided refinement loop. [28]
- **Automatic generation of formal specification and verification annotations using LLMs and test oracles**. Annotation-generation workflow for verification tools. [29]
- **Evaluating LLM-generated ACSL annotations**. Assessment of annotation quality for Frama-C-style verification workflows. [30]
- **AlphaVerus**. Self-improving verified code generation for Verus/Rust. [31]
- **RAG-Verus**. Repository-level proof synthesis for Verus with retrieval augmentation and repository context. [32]
- **Microsoft verus-proof-synthesis**. Umbrella repository for Microsoft's Verus verification automation stack. [33]
- **AutoVerus**. Automated proof generation for Rust/Verus. [34]
- **VeruSAGE**. Agent-based verification for repository-scale Rust systems. [35]
- **VeriStruct**. Planner-based verification of Verus data-structure modules. [36]
- **Generating Safety-Critical Automotive C-programs using LLMs with Formal Verification**. Joint industry-academic evaluation in a safety-critical domain. [37]
- **Clover**. Closed-loop verifiable code generation: LLM synthesises Dafny programs while a separate verifier loop checks and repairs annotations. [38]
- **SAFE**. Self-evolution approach to automated Rust/Verus proof generation, using iterative fine-tuning on successful proof traces. [39]
- **Enchanting**. Uses LLMs combined with static analysis and program verification to synthesise formal specifications from code. [40]
- **Rango**. Adaptive retrieval-augmented proof generation for Coq/Rocq: retrieves similar verified proofs to guide LLM-based proof synthesis. [41]
- **Neural Synthesis for SMT-Assisted Proof-Oriented Programming**. Fine-tunes LLMs to generate F* proof obligations dischargeable by an SMT solver, targeting proof-oriented language verification. [42]
- **Lemur**. Integrates LLMs into automated program verification pipelines (Dafny/Frama-C), combining deductive verification with LLM-generated invariants and annotations. [43]
- **Finding Inductive Loop Invariants using LLMs**. Uses LLMs to synthesise candidate loop invariants, verified by Frama-C/Boogie, with iterative feedback. [44]
- **Evaluating LLMs for VeriFast specifications**. Evaluates the ability of LLMs to generate verifiable memory-safety specifications in VeriFast for C programs. [45]
- **Can LLMs Enable Verification in Mainstream Programming?** Empirical study of LLM-assisted deductive verification across mainstream languages (Java/JML, C/Frama-C), assessing annotation correctness. [46]

6 AI-assisted mechanised proving

- **GPT-f**. Early neural theorem-proving baseline for proof search guidance. [47]
- **Draft, Sketch, and Prove**. Guides formal theorem provers using informal proof sketches. [48]
- **ProofNet**. Autoformalisation and formal proving for undergraduate-level mathematics. [49]
- **Baldur**. Whole-proof generation and repair with large language models. [50]
- **LeanDojo / ReProver**. Open tooling and datasets for Lean-based proving with retrieval-augmented models. [51]
- **COPRA**. In-context agentic theorem proving for interactive assistants. [52]
- **DeepSeek-Prover**. Open Lean prover line with synthetic-data training; the public repository tracks released models. [53, 54]
- **SubgoalXL**. Subgoal-based expert learning for theorem proving. [55]
- **LEGO-Prover**. Neural theorem proving with growing libraries. [56]
- **Lean Copilot**. LLM inference inside Lean for tactic suggestion, proof search, and premise selection. [57]
- **APOLLO**. Lean+LLM collaboration driven by compiler-guided repair. [58]
- **HybridProver**. Combines theorem proving with LLM-driven proof synthesis and repair. [59]
- **A Minimalist Proof Language for Neural Theorem Proving over Isabelle/HOL**. Proof-language design for neural theorem proving over Isabelle/HOL. [60]
- **Agentic Program Verification / AutoRocq**. Agentic Rocq/Coq verification via iterative prover feedback. [61]
- **AutoReal on seL4**. LLM-driven theorem proving in an industrial-scale verification setting. [62]
- **Vibe Coding an LLM-powered Theorem Prover / Isabellm**. Lightweight Isabelle/HOL proving with both a paper and an open implementation. [63, 64]
- **LeanNavigator**. Large-scale generation of additional Lean theorems and proofs from Mathlib4. [65]
- **AlphaProof programme**. Google DeepMind's formal mathematical reasoning effort, framed publicly through reinforcement learning and formal proof search. [66, 67]
- **Harmonic / Aristotle**. Industry effort on olympiad-level automated theorem proving and formal mathematical reasoning. [68]
- **formal-conjectures**. Repository of Lean-formalised conjectures intended to support formal reasoning research. [69]
- **TheoremLlama**. Fine-tunes general-purpose LLMs on Lean 4 proof data to create Lean-expert models. [70]
- **FVEL**. Interactive formal verification environment coupling LLMs with Isabelle/HOL for software property proving. [71]

- **Selene**. Pioneer work on automated proof generation for software verification, bridging LLMs with Lean for repository-scale proofs. [72]
- **HyperTree Proof Search**. Tree-search algorithm for neural theorem proving in Lean and Metamath, with online training via MCTS-style exploration. [73]
- **Lean-STaR**. Trains LLMs to interleave informal thinking steps with Lean proof steps, improving proof generation quality. [74]
- **REFACTOR**. Learns to identify and extract reusable lemmas from existing Lean proofs to grow the library. [75]
- **Graph2Tac**. Graph neural network approach to premise selection and tactic prediction for Lean/Coq. [76]
- **Formal Mathematics Statement Curriculum Learning**. Uses expert iteration and curriculum learning to train LLMs for formal mathematics in Lean and Metamath. [77]
- **Llemma**. Open language model specialised for mathematics, pre-trained on formal and informal mathematical text; widely used as a backbone for theorem-proving models. [78]
- **Goedel-Prover** and **Goedel-Prover-V2**. Frontier open-source Lean provers trained on large synthetic datasets, with V2 adding scaffolded data synthesis and self-correction. [79, 80]
- **DeepSeek-Prover-V2**. Advances Lean theorem proving with reinforcement learning over subgoal decomposition trees. [81]
- **Kimina-Prover**. Large formal reasoning model trained with RL, achieving strong miniF2F results. [82]
- **AlphaGeometry**. Neurosymbolic system that solves olympiad geometry problems at silver-medal level using a language model paired with a symbolic geometry engine. [83]
- **Autoformalizing Euclidean Geometry**. Demonstrates autoformalization of Euclidean geometry into Lean, translating informal geometric statements and proofs. [84]
- **LINC** and **Logic-LM** and **SatLM**. Neurosymbolic approaches that combine LLMs with first-order logic provers, symbolic solvers, and SAT/SMT reasoners respectively to achieve faithful formal reasoning. [85, 86, 87]
- **Proof Artifact Co-Training (PACT)**. Jointly trains a language model on proof artifacts (tactic states, intermediate goals) alongside informal text, improving formal proof generation across interactive proof assistants. [88]
- **Lyra**. Orchestrates dual correction in automated theorem proving: an LLM generates proof drafts while a separate refinement loop corrects both the informal reasoning and formal proof steps. [89]
- **Subgoal-based Demonstration Learning**. ICML 2024 approach that decomposes formal proofs into subgoals and uses demonstration learning to train LLMs on structured subgoal sequences for Lean theorem proving. [90]
- **HunyuanProver**. Scalable data synthesis framework combined with guided tree search for automated Lean theorem proving, achieving strong miniF2F results. [91]
- **Proving Theorems Recursively**. Decomposes complex theorems into simpler subgoals recursively, using LLMs to prove each subgoal independently and assemble the full proof. [92]

- **BFS-Prover**. Scalable best-first tree search for LLM-based automatic theorem proving in Lean, with online value estimation guiding proof search. [93]
- **InternLM2.5-StepProver**. Advances automated Lean theorem proving via expert iteration on large-scale problem sets, training step-level models with iterative data synthesis. [94]

7 Temporal logic, TLA+, model checking, and hardware formal verification

- **Guiding LLM Temporal Logic Generation with Explicit Separation of Data and Control**. Temporal-logic generation from natural-language inputs with structural guidance. [95]
- **Retrieval-Augmented TLAPS Proof Generation with Large Language Models**. Proof generation for TLAPS using retrieved verified proof fragments. [96]
- **Towards Language Model Guided TLA+ Proof Automation**. Hierarchical decomposition for TLA+ proof obligations. [97]
- **AssertLLM**. Assertion generation from design specifications and waveforms for hardware verification. [98]
- **STELLAR**. Structure-guided retrieval and generation for SystemVerilog assertions. [99]

8 Benchmarks and datasets

- **miniF2F**, **PUTNAMBENCH**, and **FormalMATH**. Benchmarks for formal mathematical reasoning and theorem proving across systems and domains. [100, 101, 102]
- **FIMO**. Challenge dataset for automated theorem proving. [103]
- **CLEVER** and **VERINA**. Benchmarks targeting verifier-in-the-loop code generation and proof-oriented programming tasks. [104, 105]
- **A benchmark for vericoding**, **AlgoVeri**, and **CASP**. Evaluation sets for formally verified program synthesis and code-level proof generation. [106, 107, 108]
- **DafnyBench** and **DafnyCOMP**. Benchmarks and competitions for LLM-assisted Dafny verification workflows. [109, 110]
- **VeriSoftBench**. Repository-scale Lean benchmark focused on software-verification obligations rather than pure mathematics. [111]
- **TLAiBench**. Benchmark repository for TLA+ specification tasks. [112]
- **AssertionBench** and **FVEval**. Benchmarks for hardware assertion generation and hardware formal-verification evaluation. [113, 114]
- **miniCodeProps**. Minimal Lean 4 benchmark for proving properties of small programs, useful for evaluating code-verification LLMs. [115]
- **Proving the Coding Interview**. Lean 4 benchmark built from interview-style algorithmic problems requiring formal verification proofs. [116]

- **Lean Workbook**. Large-scale dataset of Lean 4 problem statements collected from competition mathematics, intended to support training and evaluation of formal proving systems. [117]
- **Herald**. Benchmark for human-in-the-loop formal reasoning evaluation, measuring LLM ability to assist human provers on interactive theorem-proving tasks. [118]
- **miniCTX**. Context-dependent theorem proving benchmark for Lean that requires reasoning about file-level and project-level context beyond local tactic state. [119]

9 Adjacent specification and AI-native coding systems

- **Mog**. An AI-agent-oriented embedded language designed to be written by LLMs, with an explicit capability model and a compact language specification. [120]
- **CodeSpeak**. A spec-centric programming language and toolchain that generates conventional code from maintained specifications using LLMs. [121]
- **Spec Kit**. GitHub's open-source toolkit for spec-driven development with AI agents, centred on turning specifications into implementation plans and executable workflows. [122]

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